Tip: the journey bar is clickable - jump back to any finished stage, or between your tests, anytime.

RESULTS

Results

01 · PLS-SEM

variance-based structural model

Table 1. Measurement model

Construct / item	α	ρ_A	CR (ρ_C)	AVE	Mean	SD	Loading / weight	t	p
esg	*.86*	*.86*	*.90*	*.70*					
esg1					−0.07	1.02	.87	59.11	<.001
esg2					−0.06	1.00	.82	38.04	<.001
esg3					−0.02	0.98	.84	42.85	<.001
esg4					−0.01	1.00	.81	33.14	<.001
norm	*.88*	*.88*	*.92*	*.74*					
norm1					−0.02	1.04	.87	44.65	<.001
norm2					0.02	1.02	.84	33.03	<.001
norm3					−0.02	1.01	.86	43.36	<.001
norm4					−0.01	1.02	.86	47.10	<.001
service_quality	*.86*	*.89*	*.91*	*.71*					
service_quality1					−0.03	0.99	.84	30.18	<.001
service_quality2					−0.01	0.96	.85	33.11	<.001
service_quality3					−0.02	0.97	.86	40.17	<.001
service_quality4					0.00	1.00	.81	19.48	<.001
attitude	*.88*	*.90*	*.92*	*.74*					
attitude1					0.05	0.98	.88	55.73	<.001
attitude2					0.04	1.04	.88	59.52	<.001
attitude3					0.08	1.01	.82	31.95	<.001
attitude4					0.07	0.98	.86	47.03	<.001
intent	*.84*	*.85*	*.91*	*.76*					
intent1					−0.05	0.96	.89	78.64	<.001
intent2					−0.02	0.98	.85	55.36	<.001
intent3					−0.08	0.99	.88	76.47	<.001

Table 2. Discriminant validity — HTMT (construct × construct)

	esg	norm	service_quality	attitude	intent
esg					
norm	.11				
service_quality	.06	.14			
attitude	.05	.04	.16		
intent	.37	.26	.24	.32	

Table 3. Structural paths

H	Path	β	p	Percentile 95% CI		BC 95% CI		Result
				Lower	Upper	Lower	Upper	
H1	esg → intent	.35	<.001	.28	.43	.27	.42	Supported
H2	norm → intent	.28	<.001	.20	.36	.20	.36	Supported
H3	service_quality → intent	.20	<.001	.12	.28	.12	.27	Supported
H4	attitude → intent	.25	<.001	.17	.33	.17	.33	Supported
H5	norm*attitude → intent	.22	<.001	.15	.29	.15	.29	Supported

Table 4. Structural model quality (per endogenous construct)

Construct	R^2	R^2_{adj}	Q^2
intent	.34	.33	0.05

Table 5. Conditional effects (simple slopes)

Moderator level	β	SE	p	boot 95% CI
-1SD	.06	0.06	.290	[−.05, .19]
mean	.28	0.04	<.001	[.20, .36]
+1SD	.50	0.05	<.001	[.40, .59]

HTMT is a construct-by-construct matrix (not a per-construct value) — columns expand to the number of constructs in the model; HTMT < .85/.90 supports discriminant validity (Henseler, Ringle & Sarstedt, 2015). Structural paths carry dual bootstrap 95% CIs from ONE run — percentile (primary) and a hand-rolled bias-corrected, non-accelerated interval (the same z0-adjusted-percentile algorithm as lavaan's boot.ci.type="bca.simple", applied to seminr's raw bootstrap draws since seminr has no built-in BC option; verified to match lavaan to 6 decimals on a shared fixture). Result is Supported/Not supported from the percentile CI excluding zero (α=.05); the BC column is comparative context. R^2 and Q^2 assess the structural model (f^2: ~0.02 small, 0.15 medium, 0.35 large; Cohen, 1988). PLS-SEM deliberately has no global fit indices (CFI/TLI/RMSEA) — judge it by reliability & validity, then $R^2/Q^2/f^2$ (Hair et al., 2019). The indirect-effects table appears only when the drawn paths form a chain (X → M → Y). Formative constructs suppress AVE/HTMT and are judged by indicator weights, VIF, and redundancy convergent validity. Discriminant validity also has its own card (AVE / convergent validity); it is included here so one run gives the complete measurement-model writeup. Moderation edges (drawn on the canvas) appear as an additional structural row - an interaction construct via seminr's two-stage method (Henseler & Chin, 2010) - with the same dual bootstrap CIs as every other path. R^2(intent) = .34 (f^2 esg=0.19, norm=—, service_quality=0.06, attitude=—, norm*attitude=0.07) Bias-corrected CIs benefit from ≥7,000 resamples (Andrews & Buchinsky, 2000); consider the 10,000 publication-grade preset for final runs.

Figure. Model

– + Fit

Draw path Move Delete

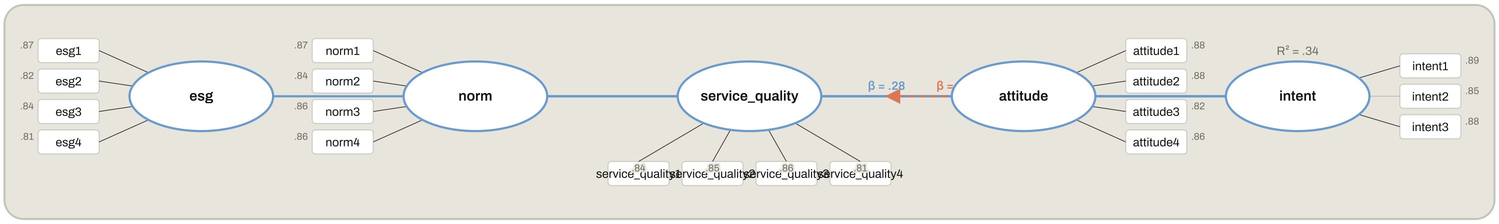
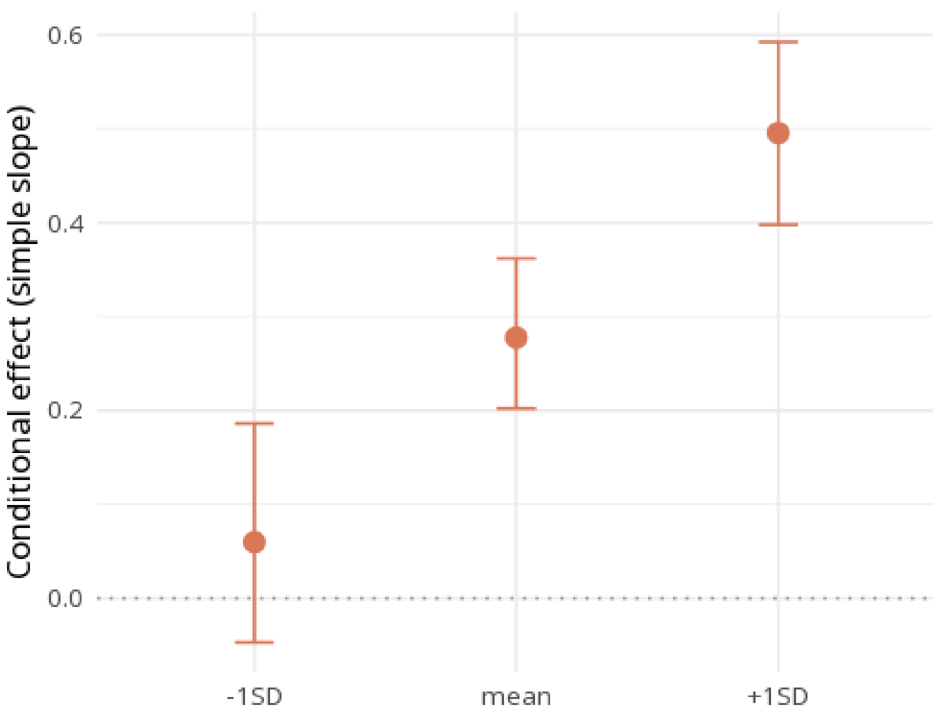


Figure. Simple slopes



Understanding the numbers

α (alpha). Cronbach's alpha for a construct - one part of the reliability & validity picture (with ρ_A, CR, AVE) PLS-SEM is judged by, since it reports no global fit indices (Hair et al., 2019). Each construct's group row carries its own α, ρ_A, CR, and AVE together (blank when that construct is modeled as formative).

ρ_A. Hair et al. (2019)'s recommended reliability coefficient for PLS-SEM constructs, reported alongside α and CR. Each construct's ρ_A is the reliability figure Hair et al. (2019) recommend leading with in PLS-SEM.

CR (ρ_C). Composite reliability for a construct - part of the same reliability & validity assessment. Each construct's CR sits alongside α, ρ_A, and AVE on its own group row.

AVE. Average variance extracted for a reflective construct; suppressed for formative constructs, which are judged instead by indicator weights, VIF, and redundancy analysis. A construct row showing AVE is reflective; a formative construct's AVE cell is blank by design.

Mean. The observed average of an indicator's raw responses, shown for descriptive context alongside its loading. Each indicator row shows its own descriptive Mean.

SD. The observed standard deviation of an indicator's raw responses. Each indicator row shows its own descriptive SD alongside its Mean.

Loading / weight. How strongly an indicator represents its construct - a loading for reflective indicators, a weight for formative ones; the column merges both since only one applies per indicator. Each indicator shows either its loading (reflective) or its weight (formative) in this single merged column.

t. The bootstrapped t-statistic testing whether an indicator's loading or weight differs from zero. Each indicator's t (with its p) tests whether its own loading/weight is reliably different from zero.

p. The bootstrapped significance of a row's estimate. Across this card's tables, p is the bootstrapped significance for that row's own estimate.

H. The hypothesis number labeling one structural path, in the order you drew it on the canvas. Each H-row is one drawn structural path, numbered in creation order.

β (path coefficient). The standardized structural path coefficient - PLS-SEM path coefficients are already on the standardized/composite scale. Each H-row's β is that path's own standardized coefficient.

Percentile 95% CI - lower bound. The lower bound of the primary bootstrap confidence interval for a structural path. Each H-row's percentile-CI lower bound, with its upper bound, is what the Result column is derived from.

Percentile 95% CI - upper bound. The upper bound of the primary bootstrap confidence interval for a structural path. Each H-row's percentile-CI upper bound completes the interval the Result column is derived from.

BC 95% CI - lower bound. The lower bound of a hand-rolled bias-corrected bootstrap interval, shown as comparative context alongside the primary percentile CI. Each H-row's BC-CI lower bound is comparative context; the Result column is always derived from the percentile CI, not this one.

BC 95% CI - upper bound. The upper bound of the same hand-rolled bias-corrected bootstrap interval. Each H-row's BC-CI upper bound completes the comparative interval alongside its lower bound.

Result. Whether a structural path's percentile CI excludes zero (Supported) or straddles it (Not supported), at $\alpha = .05$. Each H-row's Result reads directly off whether its own percentile CI excludes zero.

R^2. The share of an endogenous construct's variance explained by the paths pointing into it. Each endogenous construct's row shows its own R^2 (with R^2adj alongside); f^2 for each incoming path is reported in the note below.

R²_adj. R^2 adjusted for the number of predictor paths into that construct. Each construct's R^2adj sits beside its R^2, adjusting for the number of incoming paths.

Q^2. A measure of predictive relevance for an endogenous construct - values above zero indicate the model has predictive relevance for it (Hair et al., 2019). Each endogenous construct's Q^2 indicates its own predictive relevance.

Estimate. The size of an indirect (mediated) effect along a chained path (X → M → Y). Each indirect-effect row's Estimate is that mediated path's own bootstrapped size.

SE. The bootstrapped standard error of an estimate, used to build its confidence interval. Each row's SE feeds directly into that same row's boot 95% CI.

Boot 95% CI. The bootstrap 95% confidence interval for an indirect effect or a conditional-effect estimate. Each row's boot 95% CI is built from the same bootstrap run as its Estimate/β and SE.

Moderator level. Which point on the moderator's distribution a conditional (simple-slope) estimate is evaluated at: -1 SD, the mean, or +1 SD. Each row is the same path's slope, estimated at one moderator level (-1 SD / mean / +1 SD).

β (conditional slope). The simple-slope coefficient at one moderator level, from the conditional-effects table. Each level's β is the path's slope estimated at that specific moderator level.

How to read this test

Like CB-SEM but variance-based and prediction-oriented (good for smaller samples / formative constructs). PLS-SEM does not use CB-SEM global fit indices (CFI/TLI/RMSEA) — judge it instead by reliability & validity (CR, AVE, HTMT), then R^2, Q^2 (predictive relevance) and f^2, with SRMR the only commonly reported approximate fit index. Read the bootstrapped path coefficients (β, p); f^2 is each path's effect size (~0.02 small, 0.15 medium, 0.35 large).

APA template: In the PLS-SEM, the path from esg to intent gave β=.35, $p < .001$ (bootstrap); R^2Y=.34.

Statistical basis: PLS-SEM estimation (seminr) (Ray S, Danks N, Calero Valdéz A (2021). seminr: Domain-Specific Language for Building PLS Structural Equation Models. R package.); Discriminant validity via HTMT < .85/.90 (Henseler, J., Ringle, C. M., Sarstedt, M. (2015). "A new criterion for assessing discriminant validity in variance-based structural equation modeling." Journal of the Academy of Marketing Science, 43(1), 115-135.); f^2 effect-size benchmarks (.02/.15/.35) (Cohen, J. (1988). Statistical Power Analysis for the Behavioral Sciences (2nd ed.). Routledge.); PLS-SEM reporting standard (no global fit indices; judge by reliability/validity then $R^2/Q^2/f^2$) (Hair, J. F., Risher, J. J., Sarstedt, M., Ringle, C. M. (2019). "When to use and how to report the results of PLS-SEM." European Business Review, 31(1), 2-24.); Two-stage latent moderation via an interaction construct (Henseler, J., Chin, W. W. (2010). "A comparison of approaches for the analysis of interaction effects between latent variables using partial least squares path modeling." Structural Equation Modeling, 17(1), 82-109.); Simple slopes at -1SD/mean/+1SD (Aiken, L. S., West, S. G. (1991). Multiple Regression: Testing and Interpreting Interactions. Sage.). Full references in the exported CITATIONS.txt.